

DEPI Graduation Project Workflow

Manufacturing Downtime Data Analysis

Phase 1: Data Preparation

Responsible: Nada Waleed

- Generated a manufacturing downtime dataset using AI-assisted data generation.
- Created a dataset consisting of 4 related tables.
- Developed a Python script using Google Colab to perform data cleaning and preprocessing.
- Applied data quality checks, including handling missing values, removing duplicates, correcting data types, and standardizing records.

Deliverables:

- Raw Dataset
 - Cleaned Dataset
 - Python Cleaning Script
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Phase 2: Database Preparation

Responsible: Nada Waleed

- Established relationships between the four tables.
- Prepared the database structure using SQL Server Management Studio.
- Organized the data model to support analysis and reporting.
- Created a database backup file for deployment and future use.

Deliverables:

- Relational Data Model
 - Database Backup File
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Phase 3: Business Analysis & Dashboard Planning

Responsible: Nada Waleed & Nada Rayan

- Identified key KPIs related to manufacturing downtime analysis.
- Defined and documented 12 DAX Measures required for dashboard calculations.
- Created an HTML dashboard planning guide containing dashboard objectives, KPI mapping, visualization recommendations, and page layouts.
- Planned six dashboards to cover different analytical perspectives and business requirements.

Deliverables:

- KPI Definitions
 - DAX Measures Documentation
 - HTML Dashboard Planning Guide
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Phase 4: Dashboard Design

Responsible: Nada Rayan

- Designed six dashboard layouts using Figma.
- Created a consistent visual style across all dashboards.
- Prepared design mockups for Power BI implementation.

Deliverables:

- 6 Figma Dashboard Designs
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Phase 5: Dashboard Development

Dashboard Components

Responsible: Heba Samy

- Developed reusable dashboard components, including KPI cards, charts, filters, slicers, and navigation elements.
- Prepared components for integration into the final dashboards.

Deliverables:

- Dashboard Components Library

Dashboard Implementation

Responsible: Basmala Ahmed & Norhan Elgamel

Basmala

- OEE Overview Dashboard
- Downtime & Failure Analysis Dashboard
- Maintenance & Reliability Dashboard

Norhan

- Machine Performance Comparison Dashboard
- Quality & Defect Tracking Dashboard
- Thermal & Energy Analysis Dashboard
- Implemented the approved Figma designs in Power BI.
- Added visualizations and integrated dashboard components.
- Applied DAX measures and configured dashboard interactions.

Deliverables:

- 6 Interactive Power BI Dashboards
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Phase 6: Testing & Final Review

Responsible: Entire Team

- Reviewed data accuracy and dashboard calculations.
- Tested filters, interactions, and overall dashboard functionality.
- Ensured consistency between design requirements and implementation.
- Prepared the final presentation and project documentation.

Final Deliverables:

- Cleaned Dataset
- Database Backup File
- DAX Measures Documentation
- HTML Dashboard Planning Guide
- Figma Dashboard Designs
- Interactive Power BI Dashboards
- Final Presentation